

Qualità del Software - W10.A9

Ethical and legal issues

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The central claim of the lecture is that ethical and legal requirements for AI systems increasingly become software quality requirements. The three issues written down below are chosen because they modify quality assurance: oracle definition, test design, and post-deployment monitoring. For each I give a one-line name, a short description with its QA impact, and, where appropriate, a documented example.

Issue 1: Algorithmic discrimination without a single fairness oracle

Description. AI models trained on historical data can reproduce and amplify bias, causing disparate impact on protected groups. The hard QA point is that fairness has no single oracle: demographic parity, equalized odds and predictive parity conflict, and the impossibility theorem (Chouldechova, 2017) shows they cannot all hold when base rates differ. QA must select and justify one metric as the oracle, fix a tolerance, and document the trade-off. Legal basis: AI Act Art. 15 and GDPR.

Example. *Source: ProPublica (Angwin et al., 2016) and Chouldechova (2017), the study cited on the Fairness slide.*¹ COMPAS, the recidivism risk tool studied by Chouldechova (2017), is used by US courts. ProPublica (2016) found that, although roughly calibrated overall, it labelled black defendants high risk far more often when they did not re-offend (higher false positive rate), and white defendants low risk when they did. The vendor argued it was fair because equally calibrated. Chouldechova proved the conflict: with different base rates, equal calibration and equal error rates are incompatible; the same system thus passes one fairness oracle and fails another, so QA must make the oracle choice explicit.

Issue 2: Opaque decisions with no explanation oracle

Description. Black-box models judge people through decisions they cannot understand or contest. This is legal (GDPR Art. 22 grants a right to meaningful information about automated decisions) and a hard QA problem: the correctness of an explanation cannot be derived from the code, so no automatic oracle exists. QA must instead specify surrogates: an explainability budget (top-3 features), consistency (same input, same explanation, which is testable), and human comprehension validated by user study (over 70 percent interpret it correctly). LIME and SHAP produce explanations but do not prove they are faithful.

Example. *Source: New York State Department of Financial Services report on the Apple Card, March 2021 (complaints from November 2019).*² In November 2019 customers reported that the Apple Card (issued by Goldman Sachs) gave women far lower credit limits than their husbands despite similar or better profiles; one limit was about twenty times the spouse's. Goldman said decisions ignored gender but could not explain individual outcomes. Reviewing about 400,000 applicants, the regulator found no unlawful discrimination in 2021, yet criticised the lack of transparency that undermined trust. The case is the opacity oracle problem in practice: an automated decision that materially affected people could not be explained, reproduced or contested.

¹<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

²https://www.dfs.ny.gov/reports_and_publications/press_releases/pr202103231

Issue 3: Catastrophic failure modes and the accountability gap

Description. An AI system can be correct on average yet fail catastrophically in rare cases, and unsafe deployment choices (e.g. disabling a safety mechanism) can turn a near miss into harm. Nominal-case testing is therefore not enough: test design must target failure modes, edge cases and adversarial or out-of-distribution inputs, and AI Act Art. 15 makes pre-deployment testing and post-deployment monitoring mandatory. Accountability is also a QA concern: without complete logging (input, output, model version, timestamp, confidence) and traceability, a failure cannot be reproduced or its root cause analysed (GDPR Art. 22).

Example. *Source: course lecture (15 May) and the U.S. National Transportation Safety Board (NTSB) investigation, 2018-2019.*³ On 18 March 2018 an Uber self-driving test car killed pedestrian Elaine Herzberg as she crossed a road at night with a bicycle in Tempe, Arizona. The system detected an object about six seconds before impact but re-classified it repeatedly (unknown, vehicle, bicycle) and called for braking only 1.3 seconds before. Uber had disabled the Volvo's factory emergency braking to avoid erratic behaviour and relied on a safety driver who was not watching the road and received no alert. The NTSB found the software could not handle a pedestrian outside a crosswalk and that oversight was inadequate: untested failure modes, an unsafe trade-off and weak oversight combined.

Generative AI usage disclosure

A generative AI assistant was used to help organise and draft this summary from my lecture's notes. The factual details of the real-world cases (Uber / NTSB and Apple Card / NY DFS) were checked against public reporting. The selection of the three issues, their framing in terms of quality assurance, and the final wording are my own.

³<https://www.nts.gov/investigations/Pages/HWY18MH010.aspx>